

# Supplementary Material

## Coordinate-Explicit and Route-Specific Finite-Precision Qualification of Tangent-Only Perturbations in Coupled Finite-Element Linearizations

Bing Liang

Yangqi Ma

Weiji Sun\*

Shi He

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### Purpose, version, and scope

This supplement accompanies manuscript v0.13.0. It preserves the registered analytical checks, exact finite-element operand provenance, route-specific finite-precision error components, response-route validation, independent precision audit, effectivity anatomy, tooling lineage, and claim boundaries. The M8 and M16 cases are independent fixed finite-element linearizations, not a mesh sequence. G2-20 added synthetic response-route evaluation and a newly generated holdout, but it did not execute Abaqus, modify a finite-element operator, or import the scalable sparse-estimator results reserved for Paper 3.

## 1 Analytical verification

**Table S1:** Registered reproduction of the directional non-determination construction

| $\varepsilon$ | $\mu_J$ (matched) | $\rho_1$ (matched) | $\rho_g$ (matched) | aligned $\eta_z$ | orthogonal $\eta_z$ |
|---------------|-------------------|--------------------|--------------------|------------------|---------------------|
| 1e-12         | 1.0000e-12        | 1.0000e-12         | 1.0000e-12         | 1.0000e-12       | 0                   |
| 1e-10         | 1.0000e-10        | 1.0000e-10         | 1.0000e-10         | 1.0000e-10       | 0                   |
| 1e-8          | 1.0000e-08        | 1.0000e-08         | 1.0000e-08         | 1.0000e-08       | 0                   |
| 1e-6          | 1.0000e-06        | 1.0000e-06         | 1.0000e-06         | 1.0000e-06       | 0                   |
| 1e-4          | 1.0000e-04        | 1.0000e-04         | 9.9990e-05         | 9.9990e-05       | 0                   |
| 1e-2          | 1.0000e-02        | 1.0000e-02         | 9.9010e-03         | 9.9010e-03       | 0                   |

**Table S2:** Registered coordinate-transformation and redeclaration calculation

| $s$ | transformed $\widehat{\Delta J}_{22}$ | transformed physical $(\Delta J)_{22}$ | transformed $\eta_z$ | redeclared physical $(\Delta J)_{22}$ | redeclared $\eta_z$ |
|-----|---------------------------------------|----------------------------------------|----------------------|---------------------------------------|---------------------|
| 1   | 1.0000e-06                            | 1.0000e-06                             | 1.0000e-06           | 1.0000e-06                            | 1.0000e-06          |
| 10  | 1.0000e-05                            | 1.0000e-06                             | 1.0000e-06           | 1.0000e-07                            | 1.0000e-07          |
| 1e3 | 1.0000e-03                            | 1.0000e-06                             | 1.0000e-06           | 1.0000e-09                            | 1.0000e-09          |
| 1e6 | 1.0000e+00                            | 1.0000e-06                             | 1.0000e-06           | 1.0000e-12                            | 1.0000e-12          |

\*Corresponding author: sunweiji-1231@163.com

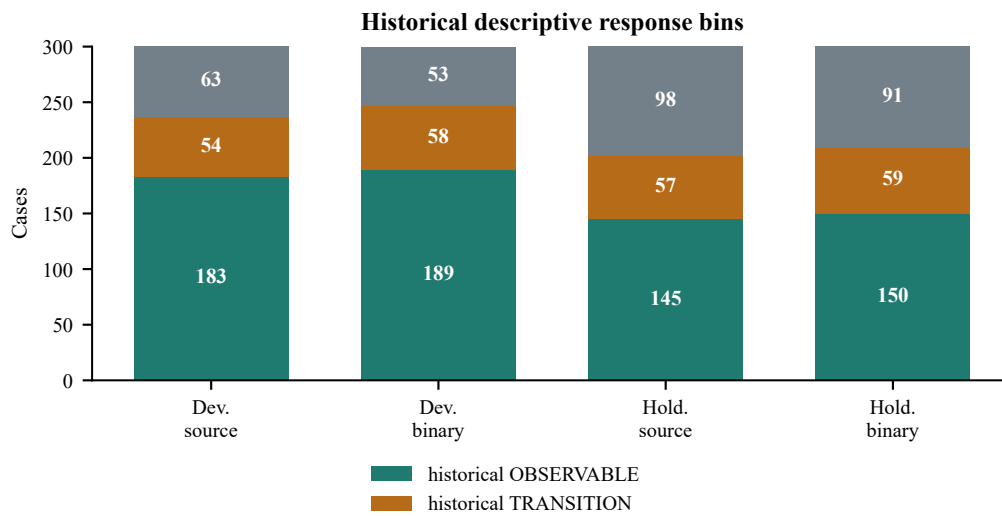
**Table S3:** Historical reproduction and direct-route coverage for the two reference problems

| Dataset/lane           | Coverage               | Severe undercoverage | Evidence role          |
|------------------------|------------------------|----------------------|------------------------|
| historical development | 287/300 (13 uncovered) | 3                    | reproduced motivation  |
| development source     | 300/300 (0 uncovered)  | 0                    | pre-specified coverage |
| development binary     | 300/300 (0 uncovered)  | 0                    | pre-specified coverage |
| holdout source         | 300/300 (0 uncovered)  | 0                    | pre-specified coverage |
| holdout binary         | 300/300 (0 uncovered)  | 0                    | pre-specified coverage |



The two source-representation markers are registered components and are not summed. This explains the historical truth-contract mismatch; it does not attribute a guard defect.

**Figure S1:** Source-representation attribution for the three historical severe-undercoverage cases. This retained diagnostic does not assign a perturbation defect



Diagnostic labels only: no validated class boundary and are excluded from confirmatory conclusions.

**Figure S2:** Historical descriptive response bins retained for provenance only. The labels have no validated class boundary and are excluded from confirmatory conclusions

## 2 Response-route theorem validation

For the response equation

$$A\delta z = -Ez, \quad A = J + E,$$

the computed route uses

$$\tilde{f} = \text{fl}(-E\tilde{z}), \quad r_\delta = \tilde{f} - A\tilde{\delta z}, \quad q_f = \tilde{f} + E\tilde{z}.$$

With majorants  $\beta_A \geq \|A^{-1}\|_2$ ,  $\alpha_E \geq \|E\|_2$ , and  $\|\tilde{z} - z\|_2 \leq U_z$ , the implemented bound is

$$U_r = \beta_A \alpha_E U_z + \beta_A \|q_f\|_2 + \beta_A \|r_\delta\|_2.$$

The development design was reused only after this route model was fixed. A new 300-system holdout then used disjoint seeds and registered condition, dimension, perturbation-amplitude, and family grids.

**Table S4:** Direct- and response-route coverage and lower-quantity counts

| Set         | Reference contract | Direct coverage | Response coverage | Lower response error | Lower response enclosure |
|-------------|--------------------|-----------------|-------------------|----------------------|--------------------------|
| Development | Decimal source     | 300/300         | 300/300           | 295/300              | 283/300                  |
| Development | Binary64 operand   | 300/300         | 300/300           | 199/300              | 229/300                  |
| New holdout | Decimal source     | 300/300         | 300/300           | 271/300              | 229/300                  |
| New holdout | Binary64 operand   | 300/300         | 300/300           | 160/300              | 188/300                  |

All 2,400 route–contract–split coverage decisions passed: 300/300 for both routes in both reference contracts on development and on the new holdout. The response equation had lower decimal-source reference error in 295/300 development and 271/300 new-holdout systems. Binary64 counts were 199/300 and 160/300. These data do not establish a universal route ranking.

## 3 Precision and implementation independence

The registered high-precision implementation was repeated at 120 and 180 decimal digits. The maximum normalized difference was  $5.16 \times 10^{-110}$  on development and  $6.82 \times 10^{-107}$  on the new holdout, with no coverage-decision change. A separately implemented Python `Decimal` backend evaluated 36 stratified holdout systems in both reference contracts. Across 72 lane comparisons, the maximum normalized difference from the registered reference was  $6.96 \times 10^{-153}$ .

The first G2-20 execution stopped because its precision comparator normalized near-zero quantities by their own magnitude. That attempt remains classified as an inconclusive reference-or-precision result. Two tooling-only corrections replaced the comparator without changing operators, thresholds, response formulas, or scientific outputs; the corrected run restarted from row 1. This lineage is reported rather than overwritten.

## 4 Effectivity anatomy and extreme ratios

For positive reference error  $e_{\text{ref}}$  and response magnitude  $d_{\text{ref}}$ , define

$$I_{\text{eff}} = \frac{U}{e_{\text{ref}}}, \quad R_U = \frac{U}{d_{\text{ref}}}.$$

For a parent bound  $U = \|A^{-1}\|_F \|r\|_2$  and exact forward error  $e = \|A^{-1}r\|_2$ ,

$$I_{\text{eff}} = \underbrace{\frac{\|A^{-1}\|_F}{\|A^{-1}\|_2}}_{\text{norm-majorant inflation}} \underbrace{\frac{\|A^{-1}\|_2 \|r\|_2}{\|A^{-1}r\|_2}}_{\text{directional inflation}}.$$

**Table S5:** Factorized direct-route effectivity anatomy

| Set              | Quantity               | Median range                            | 95th-percentile range                   |
|------------------|------------------------|-----------------------------------------|-----------------------------------------|
| Development      | Frobenius majorization | 1.0001–1.0001                           | 1.94–1.94                               |
| Development      | Directional inflation  | $1.18 \times 10^4$ – $2.31 \times 10^4$ | $3.18 \times 10^8$ – $8.53 \times 10^8$ |
| Original holdout | Frobenius majorization | 1.0050–1.0050                           | 2.69–2.69                               |
| Original holdout | Directional inflation  | $1.36 \times 10^3$ – $1.88 \times 10^3$ | $3.59 \times 10^7$ – $8.72 \times 10^7$ |

The norm-majorant median was approximately 1.0001 on development and 1.0050 on the original holdout. Directional-inflation medians instead ranged from  $1.36 \times 10^3$  to  $2.31 \times 10^4$ . Thus the central conservativeness is associated primarily with replacing the realized residual direction by a worst-direction inverse norm.

The largest finite direct-route effectivity was  $7.9321 \times 10^{86}$ , reported in the manuscript as  $\log_{10} I_{\text{eff}} = 86.90$ . It occurred for `HOLDOUT_SPD_D16_C01_E3eM17`, whose reference error was approximately  $4.9995 \times 10^{-102}$ . The extreme ratio is therefore a near-zero-denominator event, not undercoverage. Exact strings and case identifiers remain in `Paper2_extreme_ratio_record.json`.

A registered main-effects analysis gave adjusted  $R^2 = 0.869$  on development; the drop-one partial  $R^2$  for condition exponent was 0.855. The original holdout fit gave adjusted  $R^2 = 0.057$  and condition partial  $R^2 = 0.029$ . These finite-grid results support mechanism analysis within the registered design, not a universal conditioning law.

## 5 Frozen finite-element operand provenance

| Case | Dimension | Snapshot SHA-256                                                                 | Declared operand contract                                                                                            |
|------|-----------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| M8   | 196       | ACF00D823AAE<br>8BC073F14ECF<br>A322AED1B004<br>CBBA85D5FA83<br>416FEC4D24D87482 | Frozen IEEE binary64 $J, J_g, b$ , free map, and scales; transformed free coordinates; Euclidean norm.               |
| M16  | 900       | 22A51CA5BB44<br>03D1EB9499F4<br>588D6EF19CE7<br>D6C3636429D5<br>EE2FCA6AE3AB207E | Independently frozen IEEE binary64 $J, J_g, b$ , free map, and scales; transformed free coordinates; Euclidean norm. |

The M8 and M16 matrix snapshots were reproduced bitwise by independent capture witnesses. Each application reconstructed every declared binary64 operand exactly before high-precision comparison. The identity  $A = J + E$  was checked elementwise before either response route was evaluated.

**Table S6:** Rounded finite-element route values; exact decimal strings remain in the machine-readable decision files

| Case | Route        | $d_{\text{obs}}$ | $U$       | $L$       | $U/d_{\text{ref}}$ |
|------|--------------|------------------|-----------|-----------|--------------------|
| M8   | Direct       | 2.919e−12        | 5.880e−13 | 2.330e−12 | 2.523e−1           |
| M8   | Response Eq. | 2.920e−12        | 2.588e−22 | 2.920e−12 | 8.864e−11          |
| M16  | Direct       | 2.463e−11        | 3.971e−12 | 2.066e−11 | 1.922e−1           |
| M16  | Response Eq. | 2.464e−11        | 6.930e−21 | 2.464e−11 | 2.813e−10          |

## 6 Finite-element route results

## 7 Finite-element precision, residual, identity, and repeatability

**Table S7:** Rounded precision and identity checks; all gates passed

| Case | 120/180 difference      | Maximum residual        | Route-identity difference | Repeat difference |
|------|-------------------------|-------------------------|---------------------------|-------------------|
| M8   | $8.81 \times 10^{-121}$ | $2.55 \times 10^{-124}$ | $9.18 \times 10^{-123}$   | 0                 |
| M16  | $3.97 \times 10^{-121}$ | $4.14 \times 10^{-124}$ | $3.04 \times 10^{-122}$   | 0                 |

M8 required 468 s and 0.11 GiB recorded peak memory. M16 required 10.4 h and 1.35 GiB. These are offline reference costs for two fixed applications; they are not a sparse-complexity fit or an online-estimator benchmark.

## 8 Finite-element route-specific error components

**Table S8:** Component-level coverage and final route enclosure

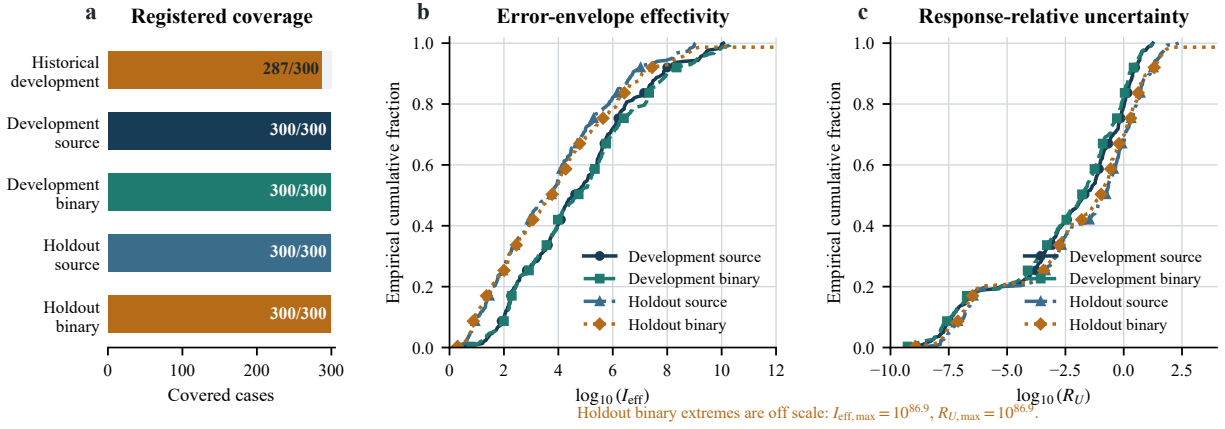
| Case | Route             | Registered components                       | Final $U$               | Coverage |
|------|-------------------|---------------------------------------------|-------------------------|----------|
| M8   | Direct difference | two parent solves; subtraction              | $5.880 \times 10^{-13}$ | 3/3      |
| M8   | Response equation | parent propagation; forcing; response solve | $2.588 \times 10^{-22}$ | 3/3      |
| M16  | Direct difference | two parent solves; subtraction              | $3.971 \times 10^{-12}$ | 3/3      |
| M16  | Response equation | parent propagation; forcing; response solve | $6.930 \times 10^{-21}$ | 3/3      |

All registered components covered their independently evaluated reference errors for the two fixed applications. Coverage is operation-graph and case specific; exact component strings remain in the archived ledgers.

## 9 Historical diagnostics retained for provenance

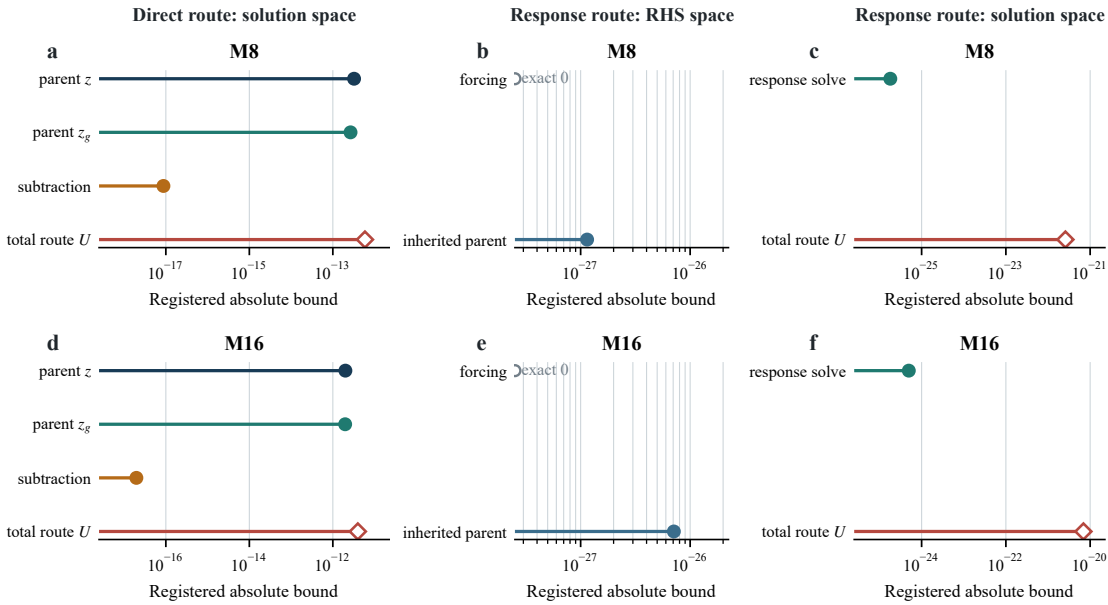
## 10 Claim boundaries and machine-readable ledger

The CSV file `Paper2_claim_evidence_matrix_v0.10.0.csv` remains the authoritative record for claims frozen through v0.10.0. The G2-20 execution report and adjudication extend it for the response-route theorem, new holdout, independent backend, and effectivity analysis; historical claim IDs and evidence tiers were not rewritten. The complete row-level matrix is distributed as CSV rather than typeset at an illegible scale in this PDF. Table S9 retains the concise inference boundary needed for human review.



Complete coverage does not imply a tight or computationally efficient uncertainty enclosure.

**Figure S3:** Historical coverage and aggregate conservativeness retained for provenance. The main manuscript replaces this summary with factorized effectivity anatomy



Component spaces are separated and are not summed across panels. The decomposition explains M8 and M16 only; it does not rank routes universally.

**Figure S4:** Historical route-component anatomy for M8 and M16. Right-hand-side and solution-space components are not added across spaces

**Table S9:** Interpretation boundary by claim domain

| <b>Domain</b>      | <b>Established</b>                                                                                                               | <b>Not established</b>                                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Theory             | Finite response identity, directional non-determination, and coordinate mapping under declared assumptions.                      | Prevalence, optimal scaling, or a universal guard rule.                                                                             |
| Synthetic evidence | Reproduction, coverage, and effectivity on the specified development and disjoint holdout designs.                               | Unrestricted-system coverage, calibrated tightness, or computational efficiency.                                                    |
| Fixed tangents     | Route-specific intervals completed separately for M8 and M16.                                                                    | Mesh law, dimension trend, or population-level generality.                                                                          |
| Route comparison   | Each route has its own uncertainty components and a case-specific decision.                                                      | Universal route ranking or superiority.                                                                                             |
| Interpretation     | Relation of $[L, H]$ to an optional threshold supplied independently by the problem definition; zero exclusion when $\tau = 0$ . | Universal threshold selection, guard safety or harm, nonlinear consequence, finite-element stability, or engineering qualification. |
| Cost               | Recorded offline cost and resource provenance for each frozen case.                                                              | Online estimator feasibility, complexity fit, or production scaling.                                                                |

## Data and code availability

The existing non-confidential reproducibility artifact is archived as Zenodo version DOI 10.5281/zenodo.21870862; the public implementation is available from the GitHub repository. The v1.1.0 G2-20 release package contains the new holdout, route results, independent-backend audit, analysis tables, scripts, and hashes. The archive excludes Abaqus executables, user-element source, solver state histories, and engineering project data.